

AI for Smarter Health Care Policy and Contract Management

Introduction

Health care organizations run on a constant stream of written content, billing and coding policies, clinical guidelines, and payer-provider contracts, much of which is dense, frequently revised, and hard to search or compare across versions, creating a real bottleneck for compliance review and policy updates as operations scale [1][2]. This report examines content management in health care as a strategic investment area for Cotiviti, defining the concept, covering relevant opportunities and threats, and closing with concrete options Cotiviti can explore.

Definition of Concepts

Content management in health care covers the organization, interpretation, and operationalization of the written that governs care delivery, billing, and payment. Billing and coding policies define how services translate into reimbursable codes and change often enough that keeping internal systems current is a constant task. Clinical practice guidelines are long, clinician-oriented documents that are hard to search for or summarize for non-clinical stakeholders who still need to reference them [3]. Payer-provider contracts are lengthy legal agreements between health care provider and a payer defining reimbursement and obligations, renegotiated periodically and requiring careful version comparisons to catch changed terms [4]. Summarization condenses any of these documents into a shorter form without losing compliance-relevant detail, while comparison of content changes means identifying exactly what changed between two versions of the same document, work that is largely manual today. Finally, converting written policy into programming language, rules, features, or models means turning plain-English policy into something a system can execute, such as a claims engine rule or a model feature, which is the step that makes a document operational and the step most prone to error and delay when done by hand [5].

Current State of Content Management

Today, most health care organizations handle billing and coding policy, clinical guidelines, and payer-provider contracts through manual review. Analysts read through documents directly, relying on basic keyword search that only helps if they already know roughly where to look, and comparing two versions of a contract or policy update is typically done line by line with little more than a document comparison tool. Converting a written policy into an executable rule for a claims engine or a model feature is a separate manual translation step, handled by a different team that must re-read the policy and decide how to encode it [6].

This process is slow and inconsistent. Two analysts can interpret ambiguous language differently, and a missed update during a busy review cycle can leave a claims system running outdated rules for weeks before anyone notices [7]. As document volume grows and regulations change more often, this manual approach becomes the bottleneck limiting how quickly a payer responds to policy change or catches a costly contract discrepancy [8].

New Opportunities with Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation (RAG) change this picture by letting analysts ask plain English questions directly against a policy or contracts corpus and get grounded answers pulled from the actual

source text, instead of manually searching for the right section [9]. The same retrieval setup can power automated comparison between the document versions, surfacing what changed in a policy or contract update in seconds rather than through a line-by-line manual read [10]. A fine-tuned or well-prompted model can also assist with converting written policy into structured rules or model features by drafting a first pass translation for a human reviewer to confirm, cutting down the time between policy update and its implementation in the production system [11].

Threats and Obstacles

The biggest risk is exposing protected health information (PHI), since policy examples, contract details, and clinical guidance can reference patient or member data, and that risk extends to the AI pipeline itself if PHI is not masked before it reaches a model. Cloud platforms such as Amazon Bedrock and Azure AI Foundry reduce this risk through building guardrails and HIPAA eligible infrastructure under a signed Business Associate Agreement, but the platform alone does not guarantee compliance, the organization is still responsible for configuration, access control, and how PHI is handled before and after each model call [12][13]. A second obstacle is data quality, since many payers are still digitizing decades old contracts and policy documents, and Optical Character Recognition (OCR) accuracy on these scans is often inconsistent due to faded text and irregular formatting, which limits how reliable any downstream LLM pipeline can be [14]. Finally, hallucination remains a concern in a regulated domain like this, since an LLM that confidently states an incorrect rule or misreads a contract term could lead to a costly compliance error if a human reviewer is not kept in the loop [15].

Proposed Solution

I propose Cotiviti invest in a retrieval augmented generation pipeline built on a cloud platform such as Amazon Bedrock or Azure AI Foundry, design specifically for billing and coding policy, clinical guidelines, and payer-provider contracts. The pipeline would ingest documents into a vector store, with a PHI masking step applied before ingestion so sensitive identifiers never reach the model, and a second masking passes on model output before it reaches a human reviewer. On top of this, the system would offer two core capabilities, a question-and-answer interface that lets analysts query policy or contract content in plain English with answers grounded in the retrieved source text, and an automated document comparison feature that flags exactly what changed between two versions of a policy or contract. A human reviewer would remain in the loop for any output used in a compliance decision or claims rule update, treating the model's response as a draft to verify rather than a final answer.

A Note on Proposed Solution and Methods

The proposed solution in this report comes from my own studies, professional experience, and education in AI and health care technology, with inspiration drawn from my work as an AI Engineer at Worcester Polytechnic Institute, where I built a fine-tuned LLM and retrieval augmented generation system for a resource-sharing platform for non-profit organizations. This proposal also draws in part on my experience as an IT Intern at Fallon Health, where I designed and deployed a multi-agent LLM framework for document-based Q&A that improved informational access by 45 percent in the organization. I used Claude, an AI assistant by Anthropic, to assist with research, organize the report into sections, and rephrase parts of the writing for clarity within the two-page constraint, with sources verified and cited as instructed in the prompt.

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